Detailed Topology Scan Report - Statistics

Total scan points: 27

Phase counts: trivial=12, TI=6, HOTI=9
gap range: [-2.000e+00, 2.000e+00]
|chern| max: 6.241e-16
|Wilson_loop| range: [0.000e+00, 3.615e+00]
Z2=1 count: 15
edge_state=1 count: 15
corner_state=1 count: 0

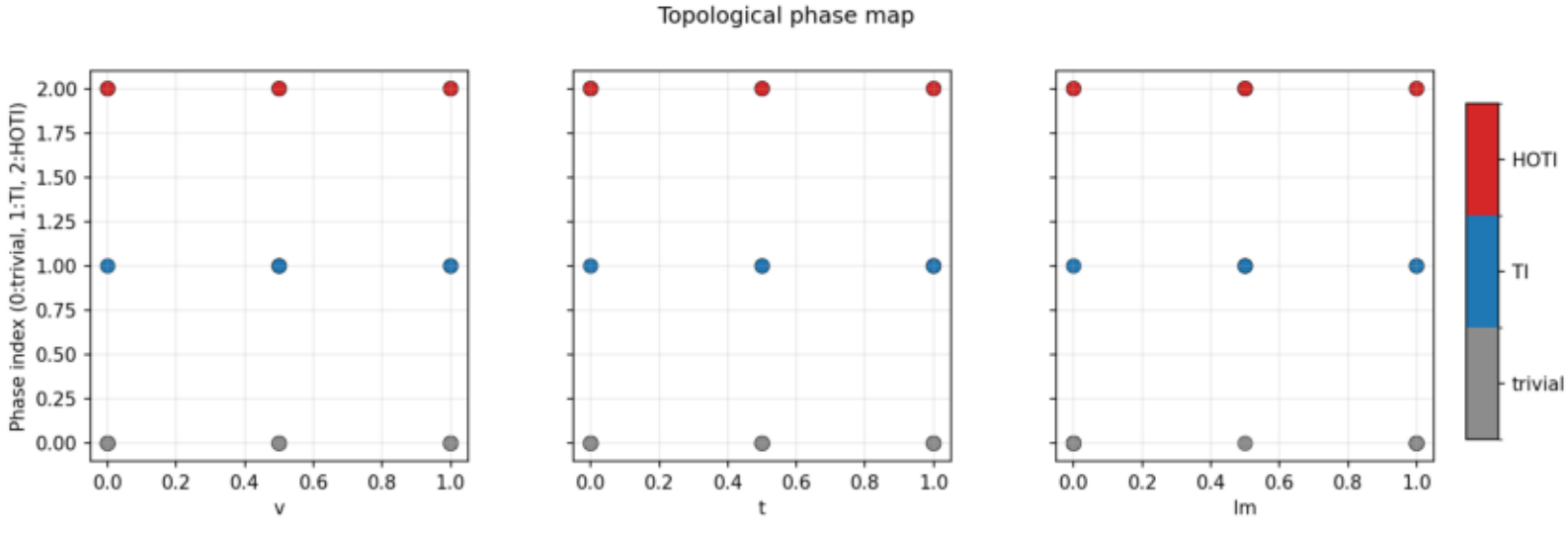
Parameter dependence (phase-count by each axis):
v=0.0: trivial=5, TI=1, HOTI=3
v=0.5: trivial=3, TI=3, HOTI=3
v=1.0: trivial=4, TI=2, HOTI=3
t=0.0: trivial=4, TI=1, HOTI=4
t=0.5: trivial=4, TI=2, HOTI=3
t=1.0: trivial=4, TI=3, HOTI=2
lm=0.0: trivial=6, TI=1, HOTI=2
lm=0.5: trivial=1, TI=3, HOTI=5
lm=1.0: trivial=5, TI=2, HOTI=2

Detailed Topology Scan Report - Candidate Points

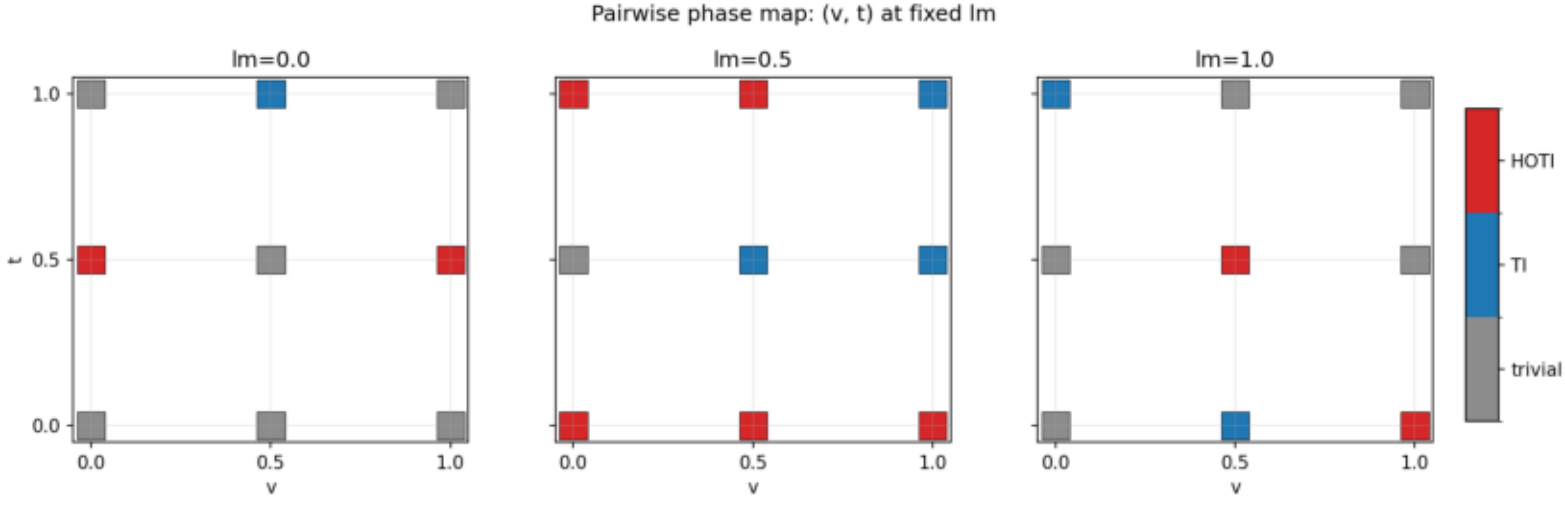
Top 10 smallest-|gap| points (potential transition candidates):

01. v=0.5, t=0.5, lm=1.0, gap=5.551e-17, chern=2.120e-16, W=-1.058, Z2=1, edge=1, co
02. v=0.5, t=1.0, lm=0.5, gap=-5.551e-17, chern=3.534e-16, W=-0.605, Z2=1, edge=1, c
03. v=1.0, t=1.0, lm=1.0, gap=1.237e-16, chern=0.000e+00, W=-0.263, Z2=0, edge=1, co
04. v=1.0, t=0.0, lm=0.0, gap=2.449e-16, chern=1.548e-49, W=0.000, Z2=0, edge=1, cor
05. v=0.0, t=0.5, lm=0.5, gap=-2.680e-16, chern=1.414e-16, W=0.293, Z2=0, edge=1, co
06. v=1.0, t=0.0, lm=0.5, gap=-4.161e-16, chern=3.878e-17, W=1.360, Z2=1, edge=1, co
07. v=0.0, t=0.5, lm=1.0, gap=-6.012e-16, chern=4.240e-16, W=-0.200, Z2=0, edge=1, c
08. v=1.0, t=0.5, lm=0.0, gap=-6.021e-16, chern=6.367e-17, W=-0.962, Z2=1, edge=1, c
09. v=0.0, t=0.0, lm=0.5, gap=-7.364e-16, chern=-6.703e-17, W=0.533, Z2=1, edge=1, c
10. v=0.0, t=1.0, lm=0.5, gap=-8.094e-16, chern=-2.119e-16, W=-0.861, Z2=1, edge=1,

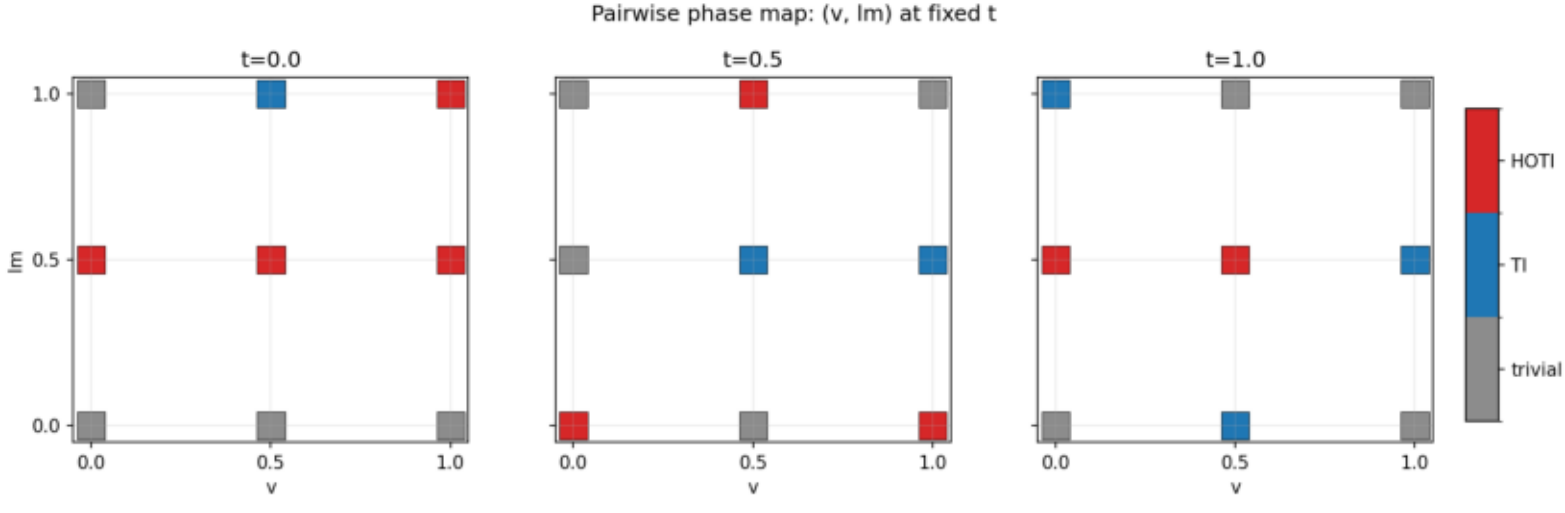
topology_phase_map.png



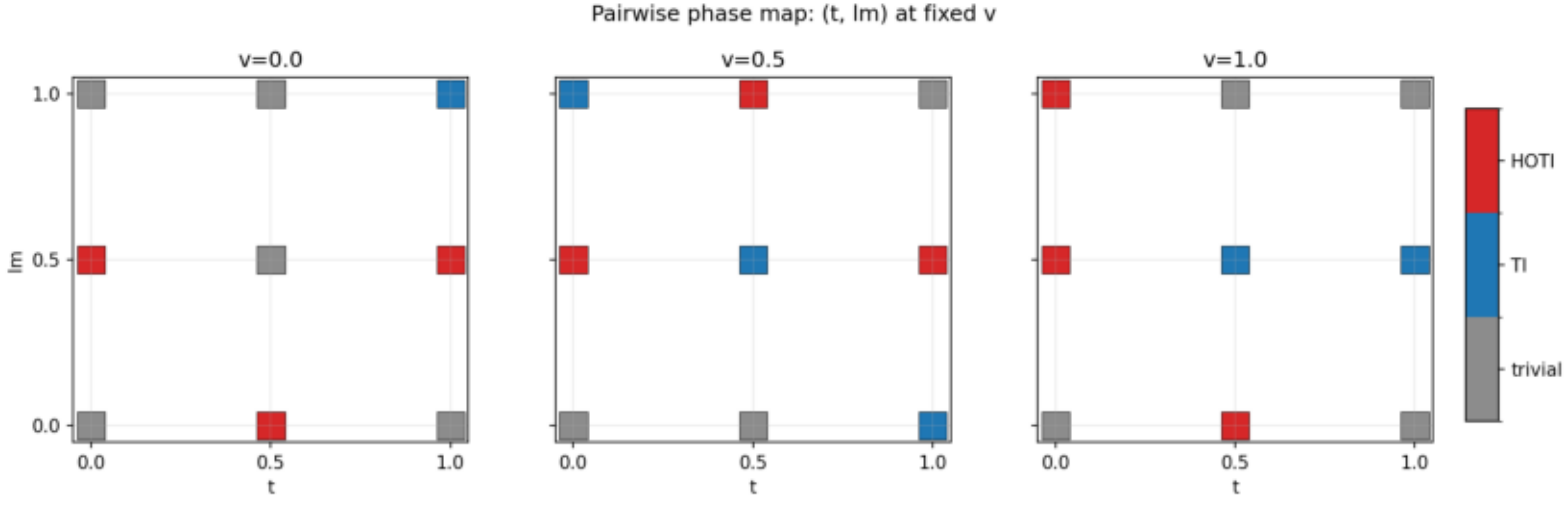
phase_map_v_t_by_lm.png



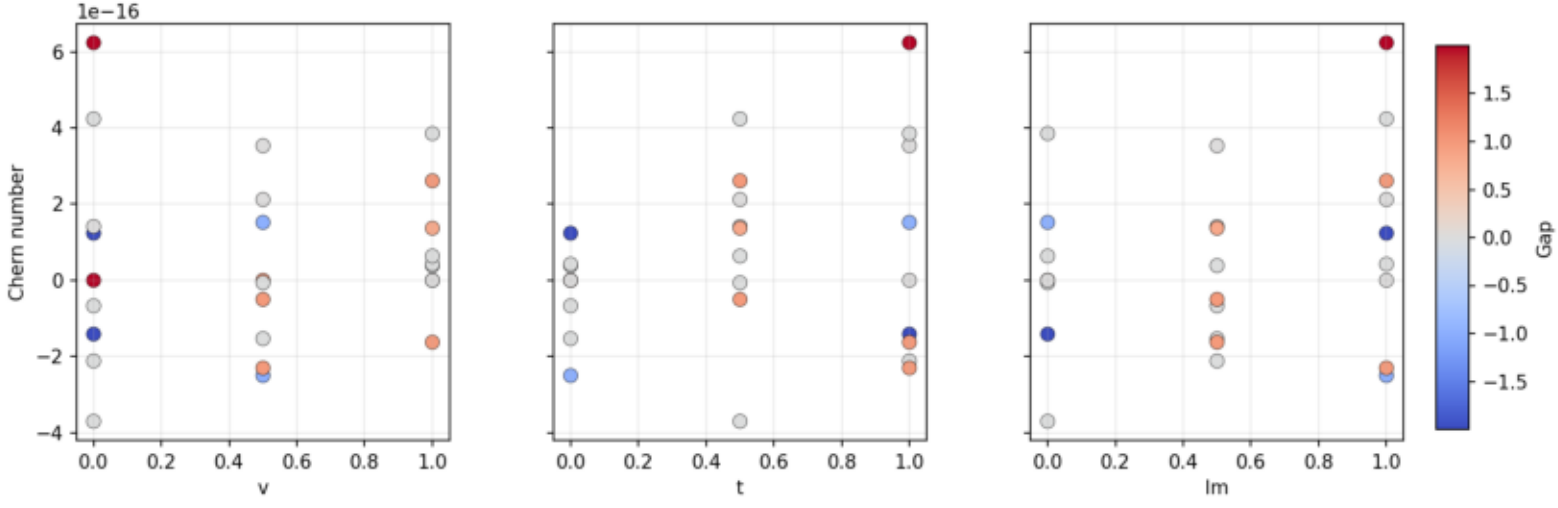
phase_map_v_lm_by_t.png



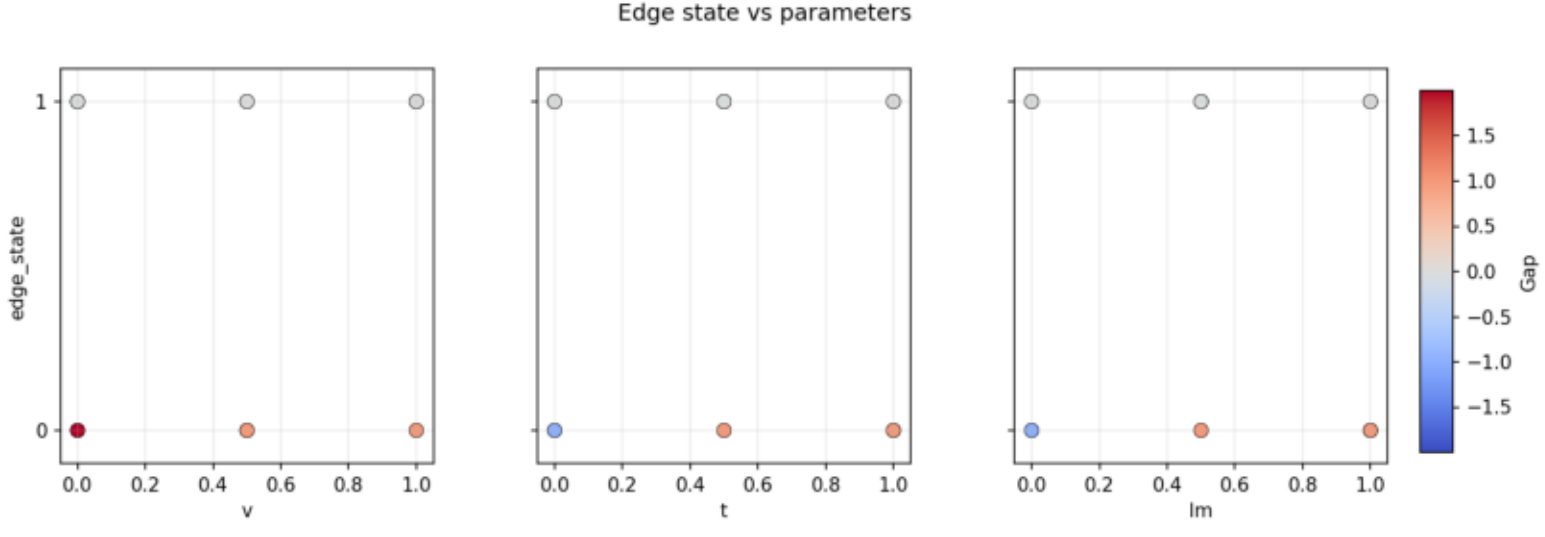
phase_map_t_lm_by_v.png



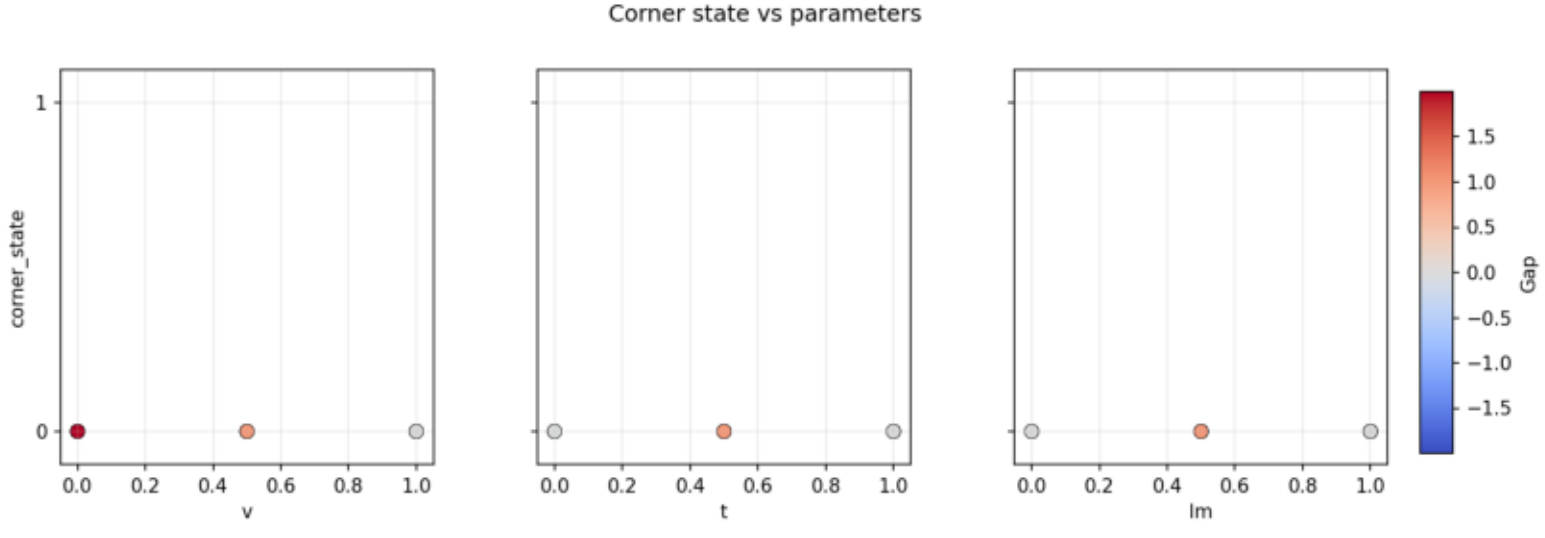
chern_vs_parameters.png



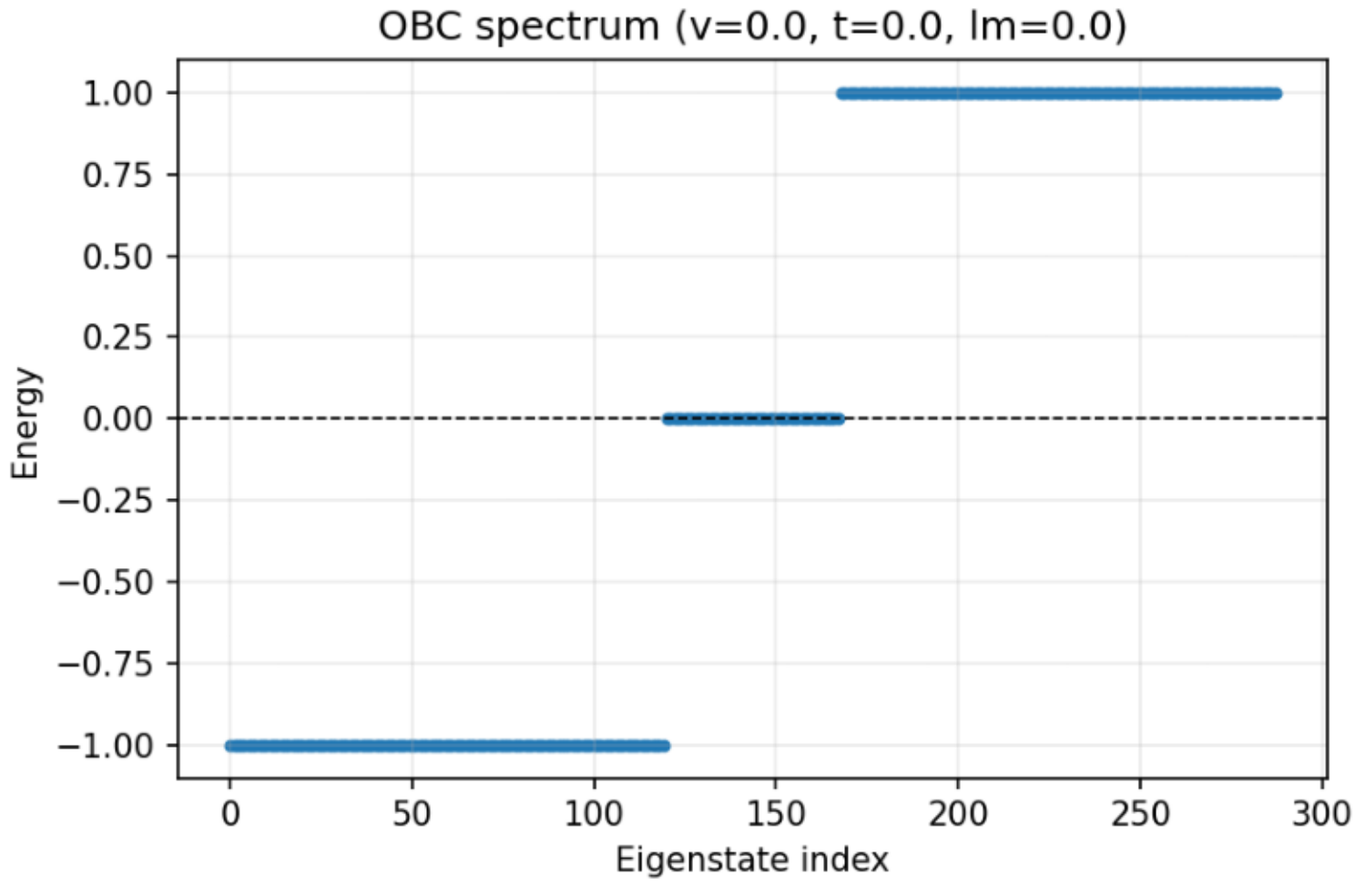
edge_state_vs_parameters.png



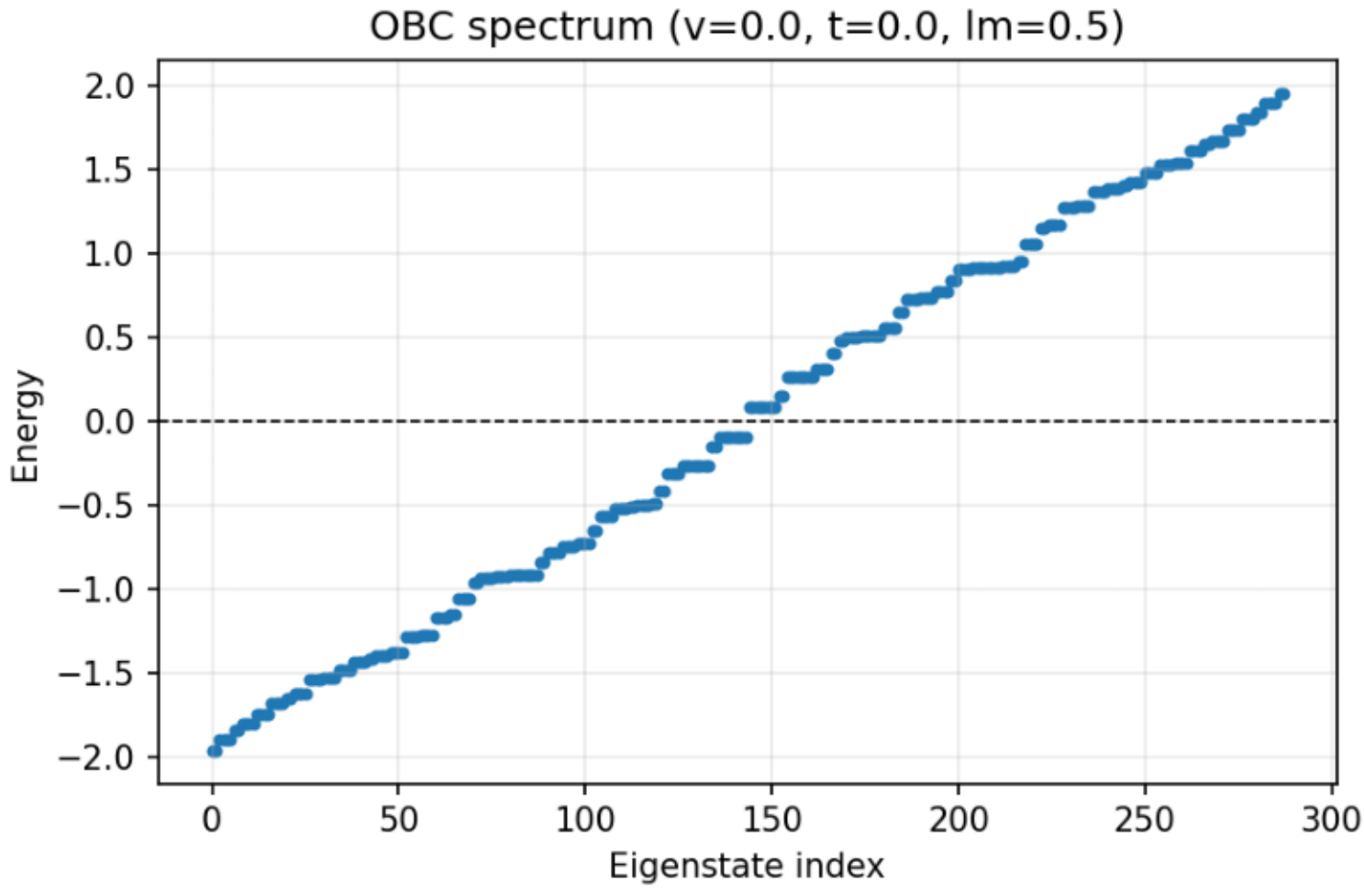
corner_state_vs_parameters.png



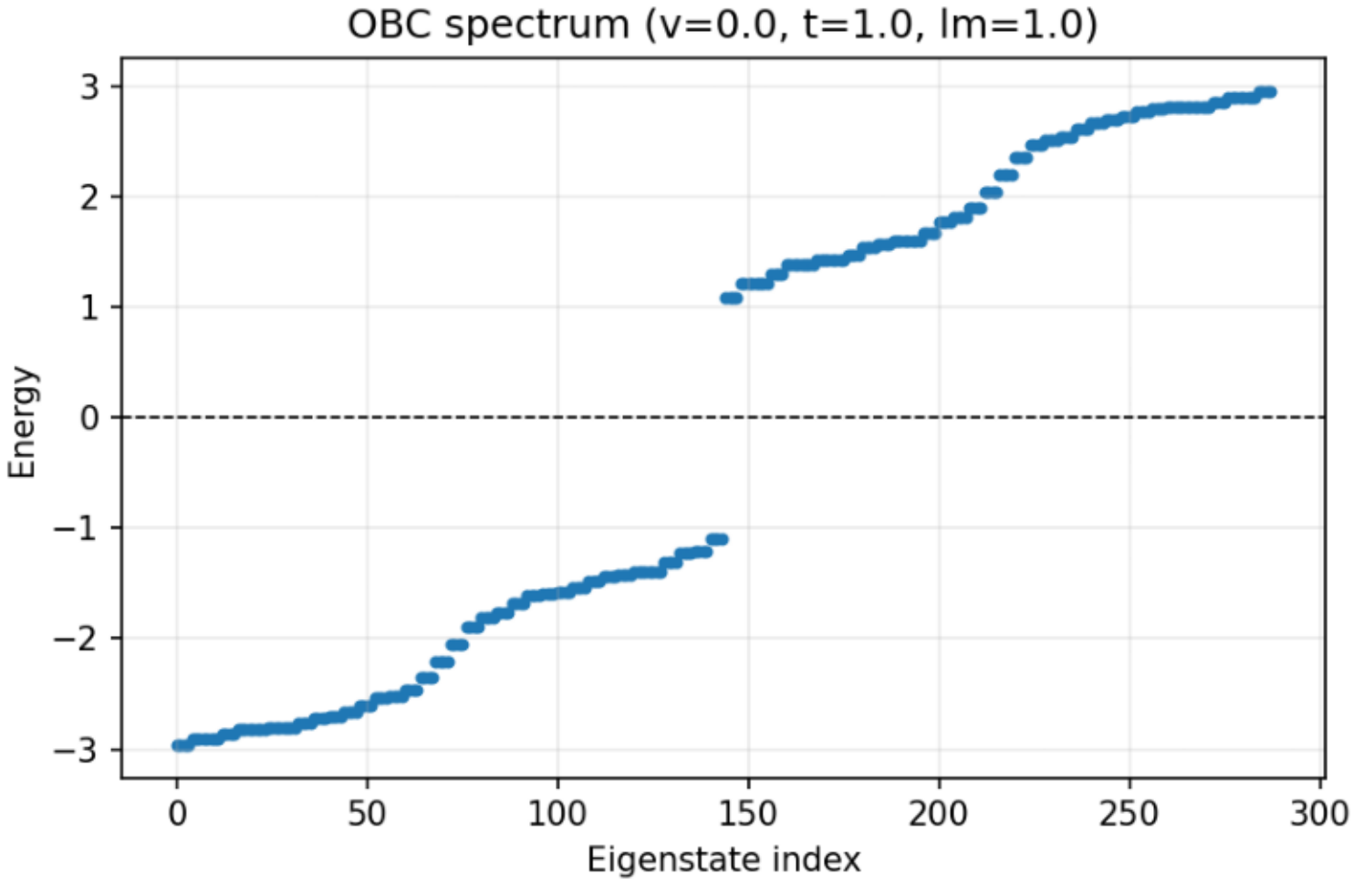
obc_spectrum_v0.0_t0.0_lm0.0.png



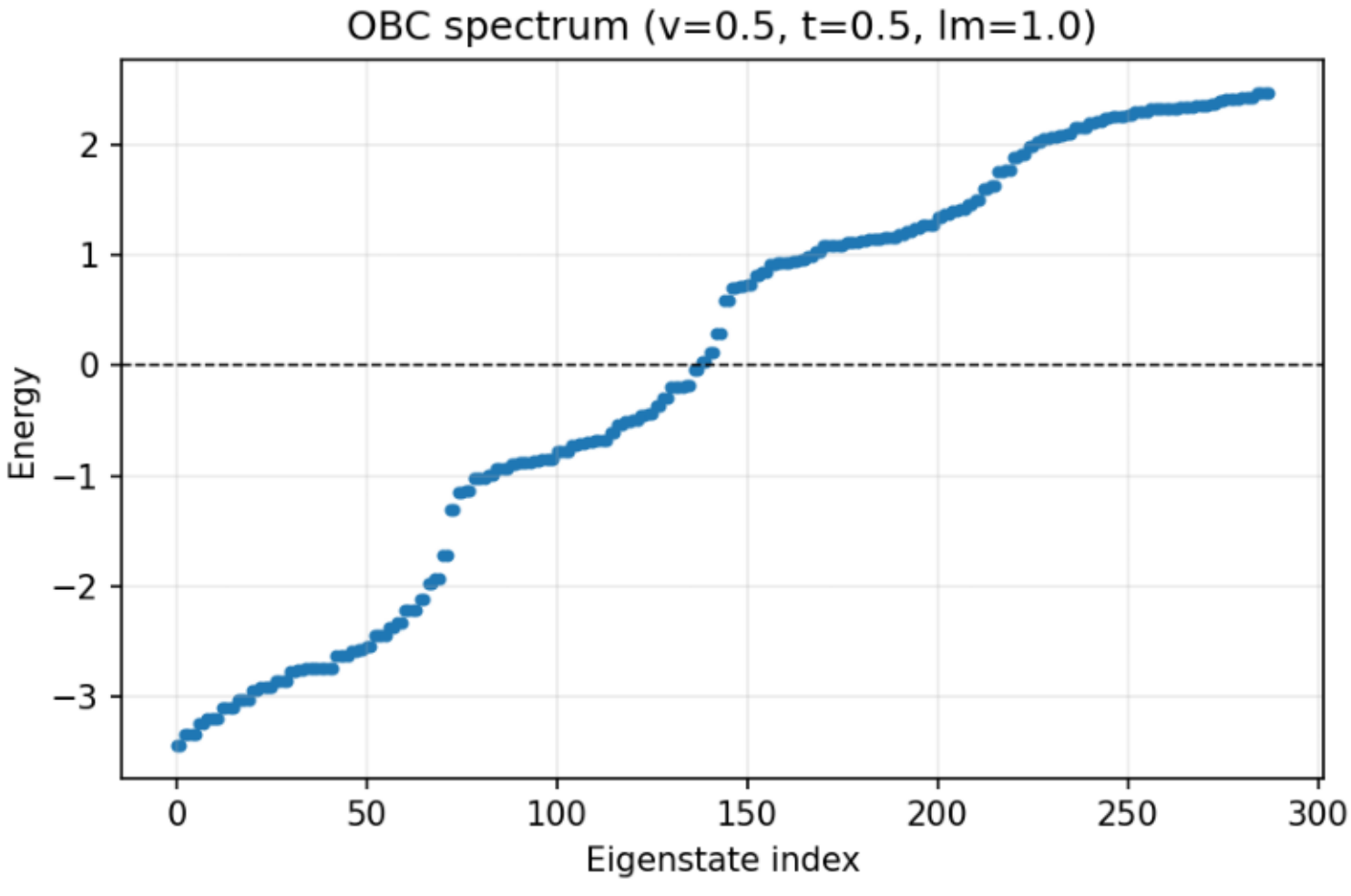
obc_spectrum_v0.0_t0.0_lm0.5.png



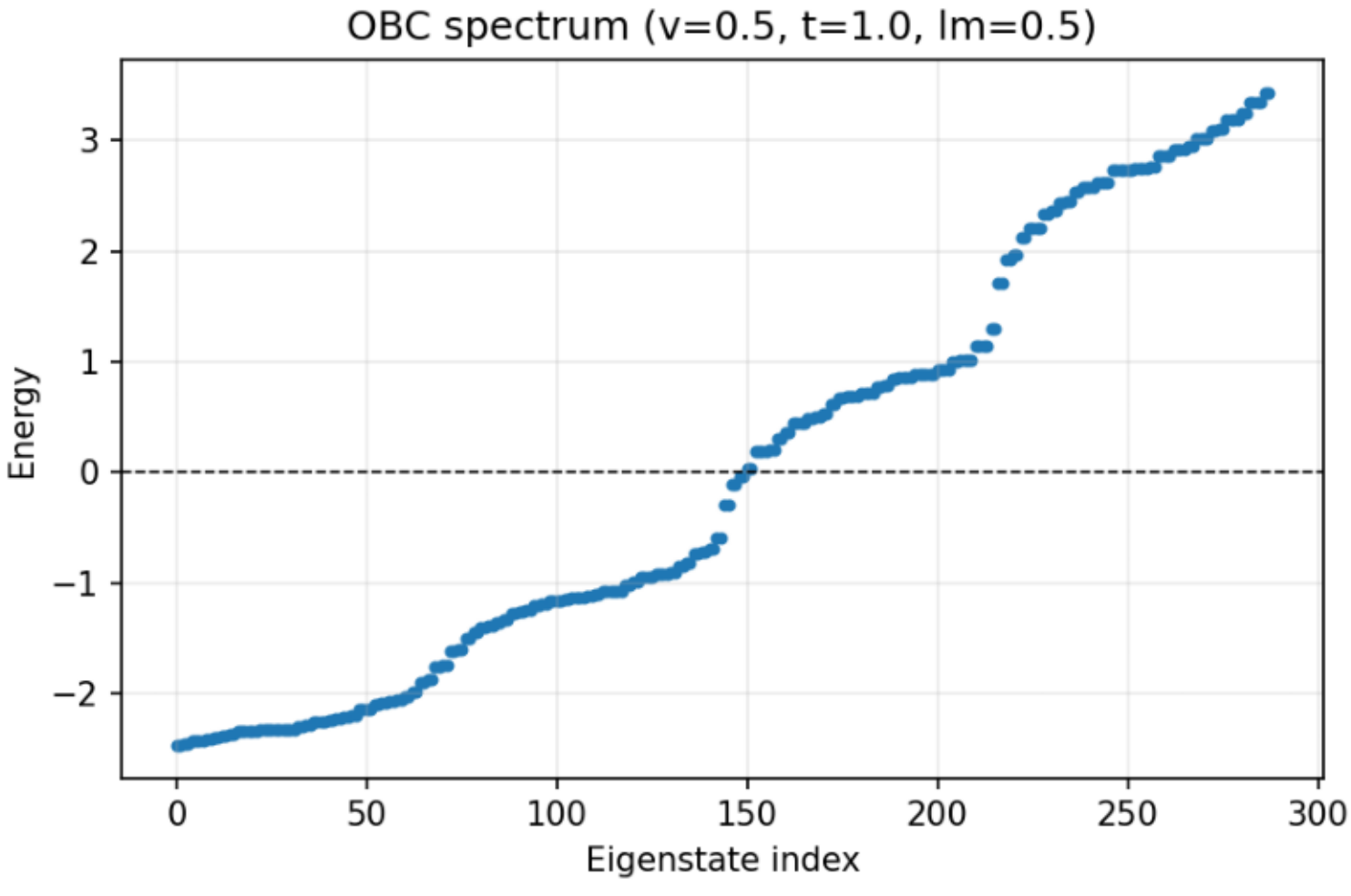
obc_spectrum_v0.0_t1.0_lm1.0.png



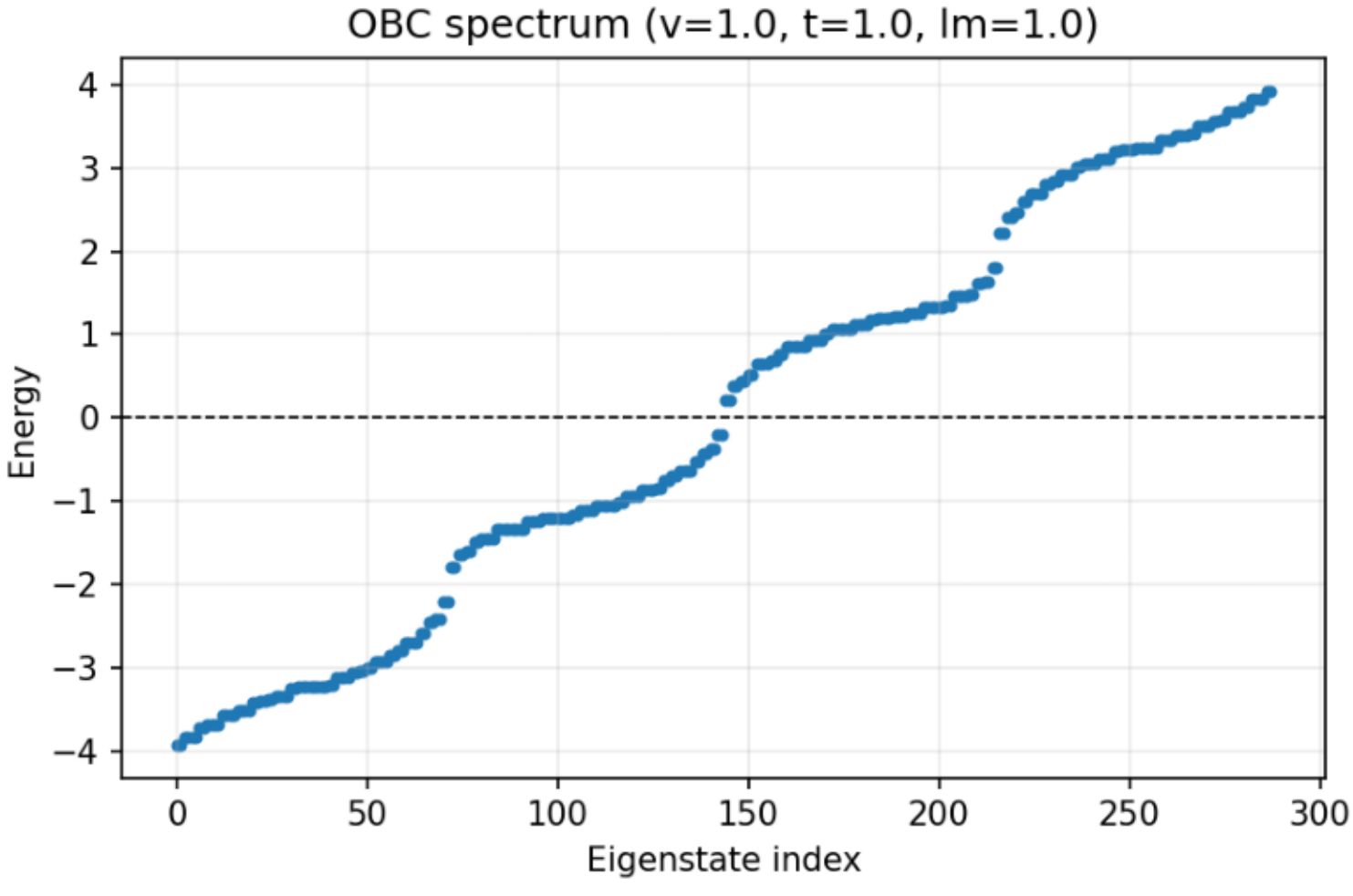
obc_spectrum_v0.5_t0.5_lm1.0.png



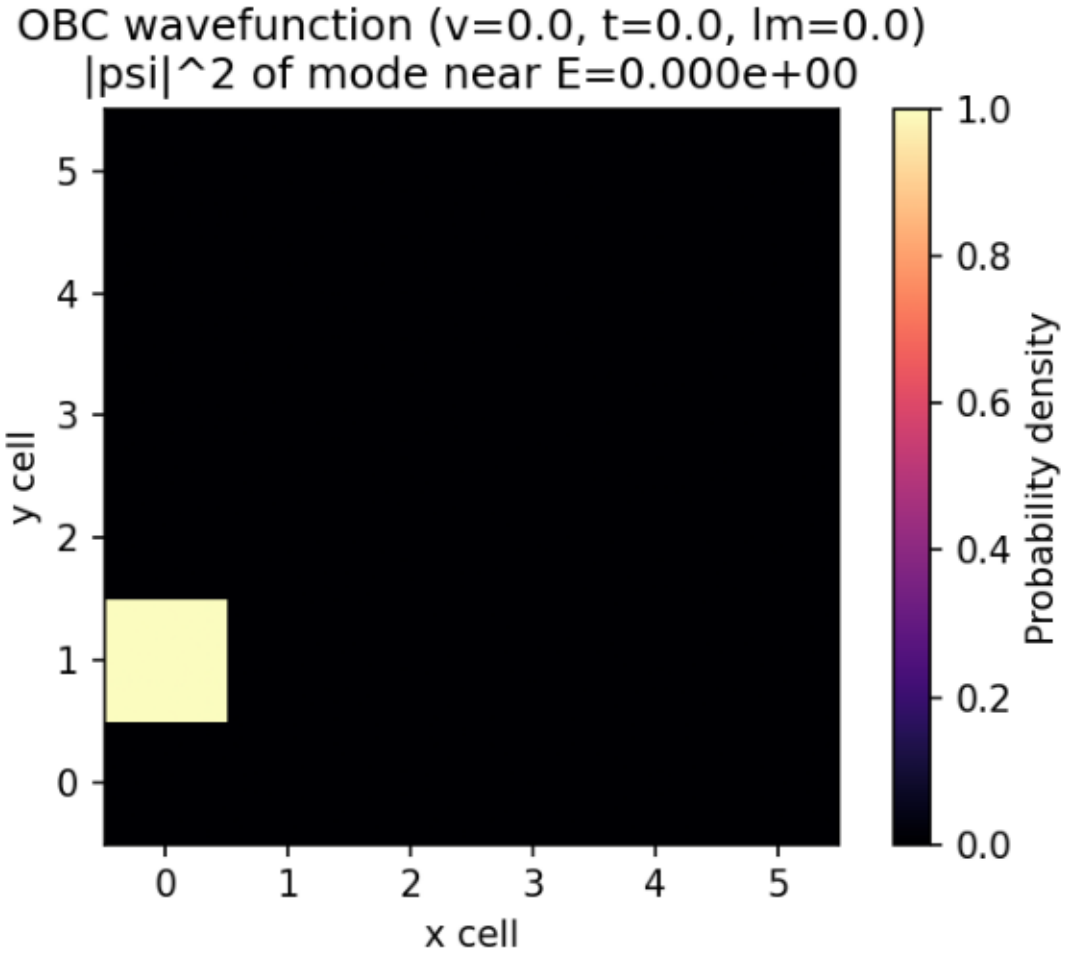
obc_spectrum_v0.5_t1.0_lm0.5.png



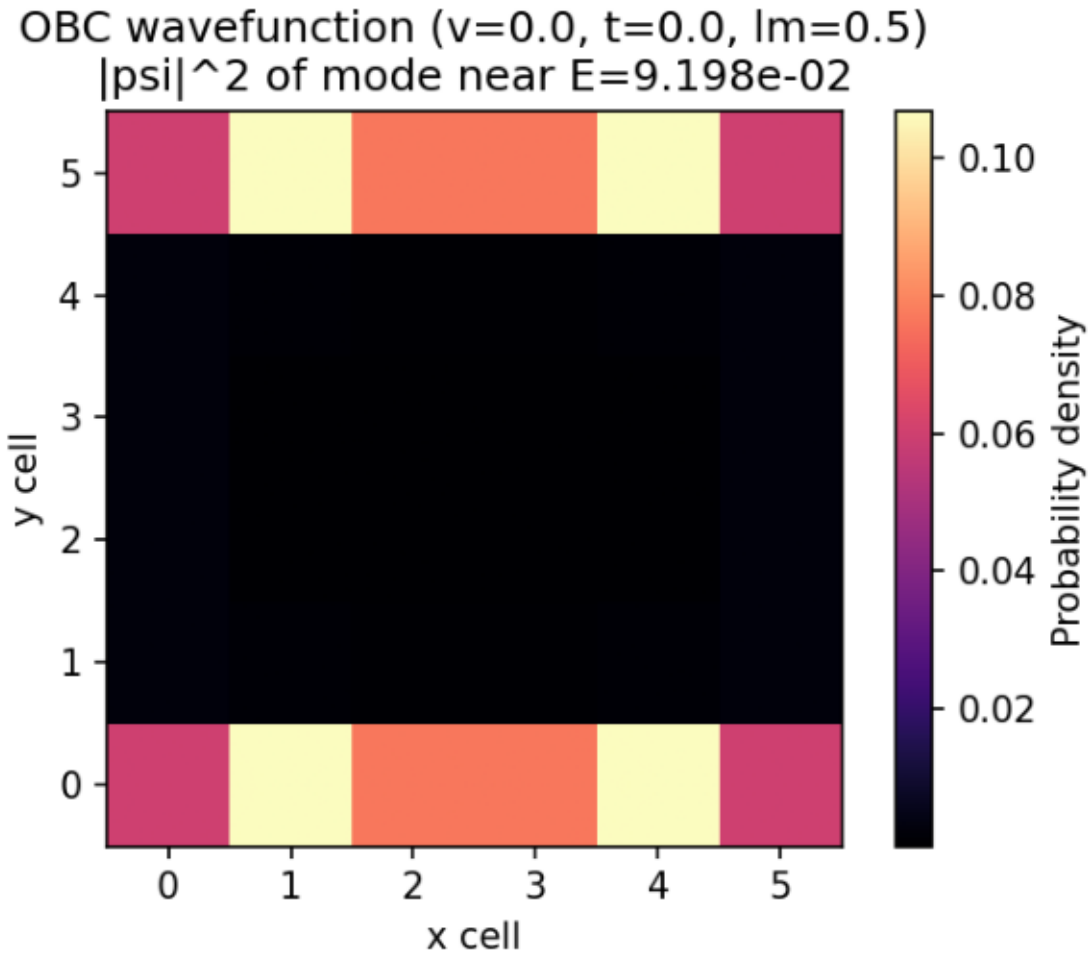
obc_spectrum_v1.0_t1.0_lm1.0.png



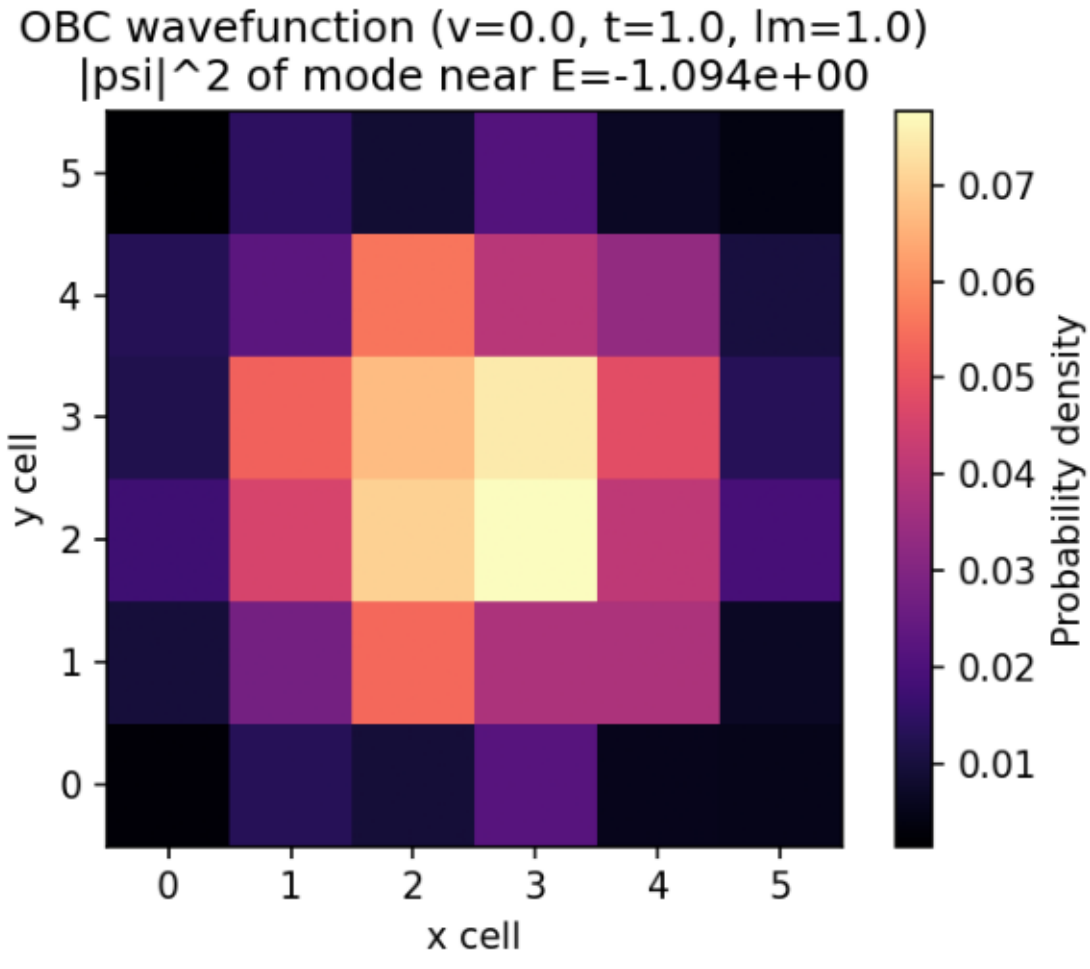
obc_wavefunc_v0.0_t0.0_lm0.0.png



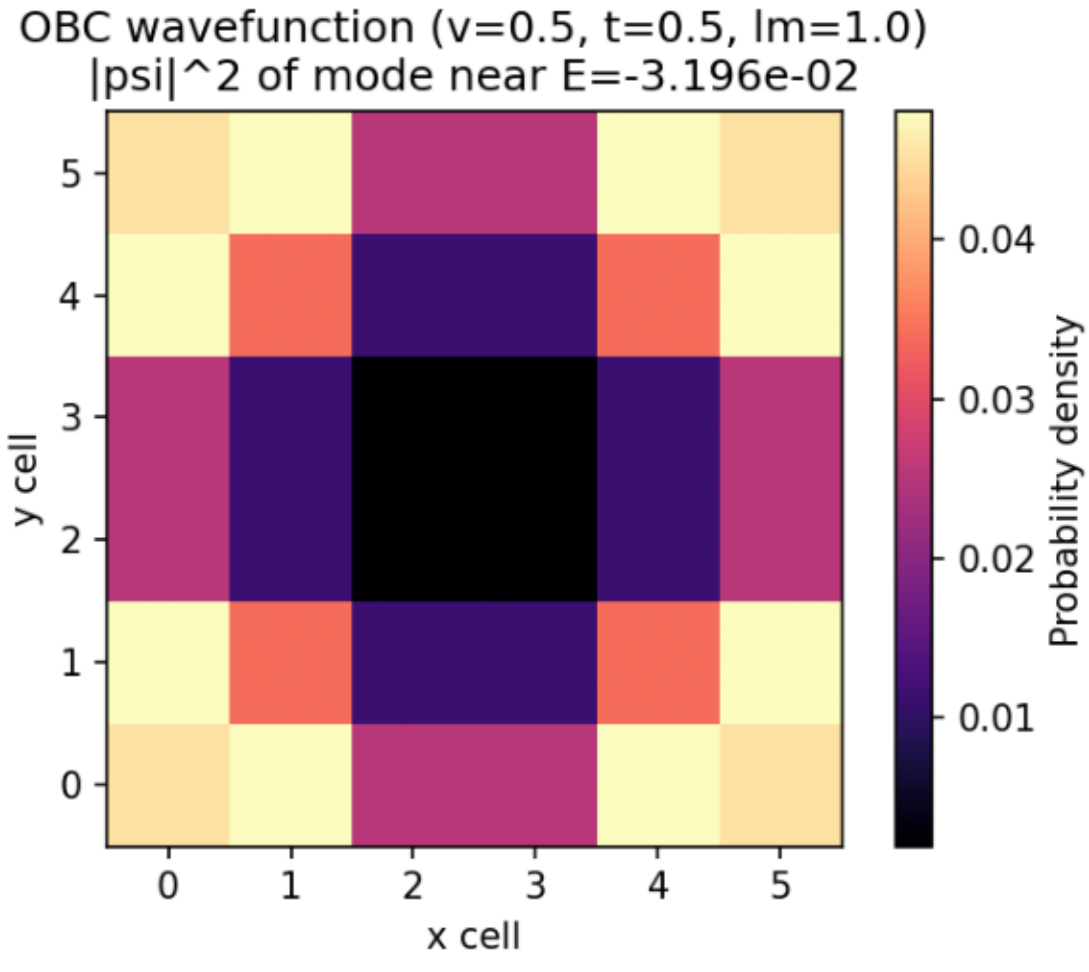
obc_wavefunc_v0.0_t0.0_lm0.5.png



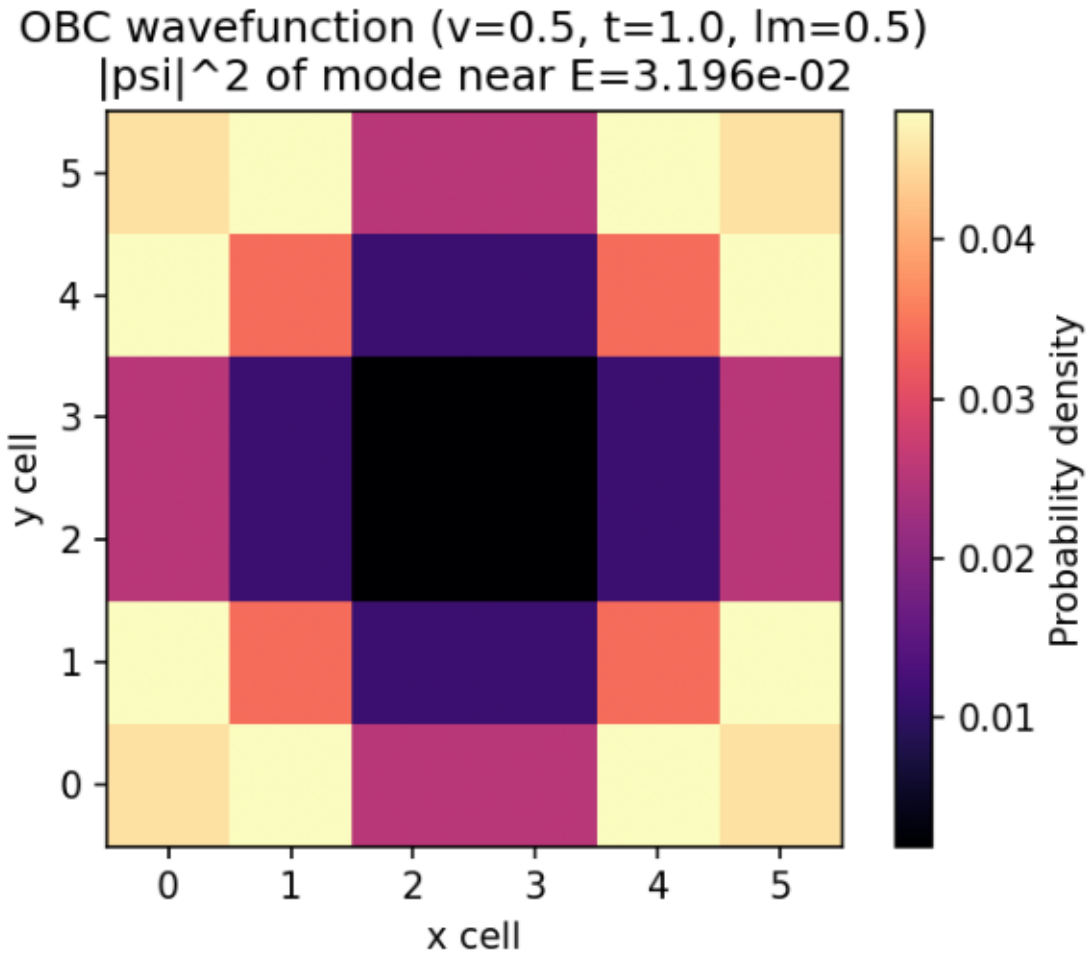
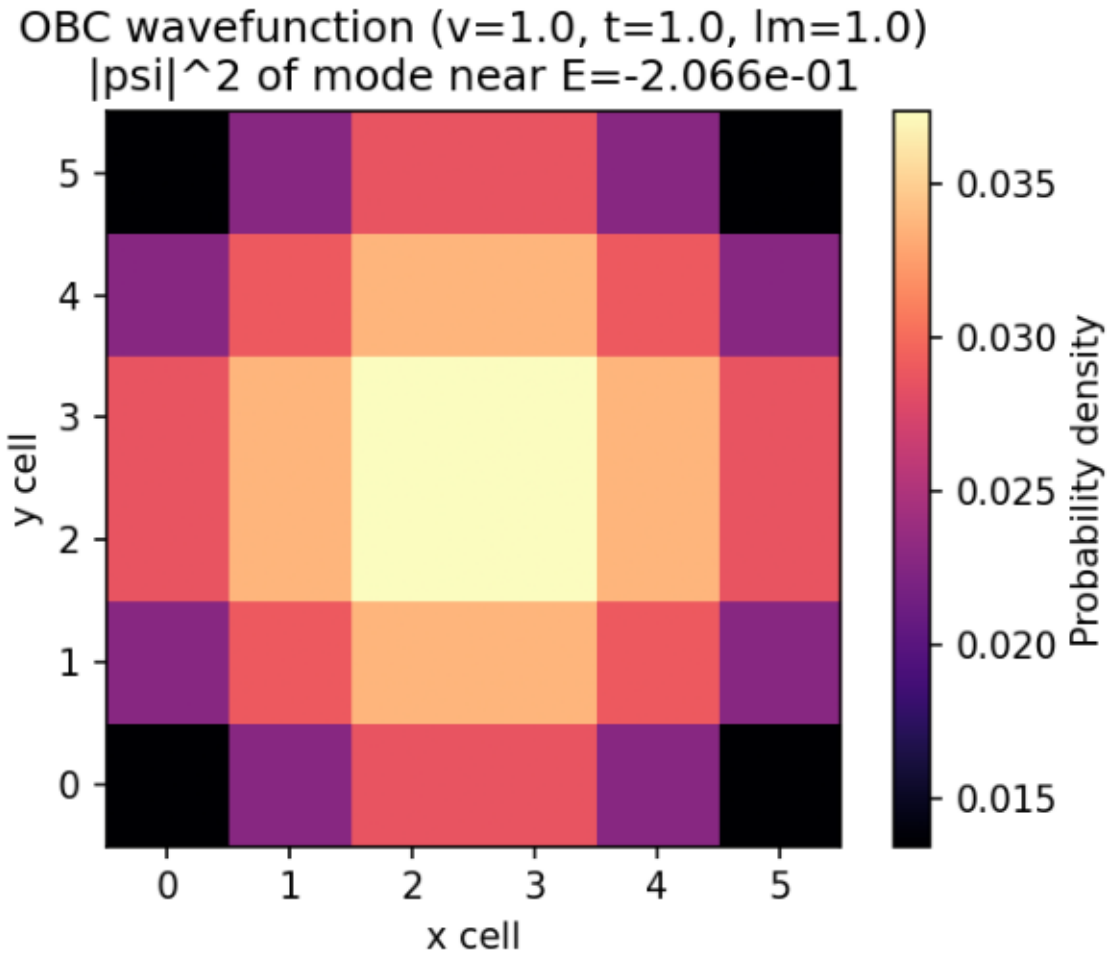
obc_wavefunc_v0.0_t1.0_lm1.0.png



obc_wavefunc_v0.5_t0.5_lm1.0.png



obc_wavefunc_v0.5_t1.0_lm0.5.png



obc_wavefunc_v1.0_t1.0_lm1.0.png